NyaySaakshya

Cryptographic Security Posture Assessment

```
Assessment 645090e0bc44ba9d
Capture ce5377348e22ad92c33d705e31388944bad8224534d478c38b57a75d3ba4a501
Run 73ef94a35c4a442d8ed7d0f762175c8f
Analysed 2026-10-03T04:36:38Z
```

OVERALL POSTURE	POSTURE SCORE	EVIDENCE COVERAGE	ML LANE
CRITICAL	44 / 100	100.0% (1 of 1 sessions assessed)	Disabled

1. Executive summary

Overall posture	CRITICAL
Posture score	44 / 100 F2-group-damped
Evidence coverage	100.0% (1 of 1 sessions assessed)
Protocols observed	smtp
Issue groups	10 total · 2 penalising the score
Abstentions	2
ML lane	Disabled

1 session(s) were reconstructed from the capture; 1 were assessed and 0 were abstained from.

2 issue group(s) penalise the posture score. The highest severity recorded is HIGH.

2. Assessment scope

What this assessment covers, and what it structurally cannot.

Analysis type	Passive PCAP analysis
Capture identifier (SHA-256)	ce5377348e22ad92c33d705e31388944bad8224534d478c38b57a75d3ba4a501
Analysis timestamp	2026-10-03T04:36:38Z time of analysis, not of report generation

NyaySaakshya observes traffic that was already captured. It does not connect to any server, hold any private key, decrypt any session, or read message content. Anything that cannot be determined from the captured bytes is reported as not observable rather than inferred.

Sessions per protocol

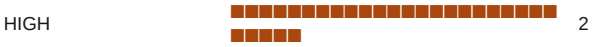
PROTOCOL	SESSIONS
smtp	1

3. Overall security posture

Posture	CRITICAL
Score	44 / 100
Evidence coverage	100.0% (1 of 1 sessions assessed)
Scoring formula	F2-group-damped

100 minus 56.00 across 2 issue group(s) under F2-group-damped

The score starts at 100 and each penalty above is subtracted from it. Only observed issues penalise; compliant observations earn no credit, and missing evidence neither penalises nor rewards.



Score decomposition

ISSUE CLASS	SEVERITY	RECURRENCE	WEIGHT	RECURRENCE MULTIPLIER	PENALTY
Certificate key strength	HIGH	1	28	1	-28
Certificate signature algorithm	HIGH	1	28	1	-28

The severity weights and band thresholds behind this score are transparent engineering policy, not a calibrated measurement derived from incident data.

4. Evidence coverage

How much of the captured traffic the analysis could actually assess.

Sessions total	1
Sessions assessed	1
Sessions abstained	0
Assessed fraction	100.0%

Coverage is reported beside the posture score and is never folded into it. A score computed over a small share of the traffic and the same score computed over nearly all of it are different claims.

Evidence states observed across all fields

EVIDENCE STATE	COUNT	SHARE
AMBIGUOUS	1	12.5%
NOT OBSERVABLE	4	50.0%
OBSERVED	3	37.5%

Session completeness

COMPLETENESS	SESSIONS
COMPLETE	1

5. Protocol posture

Posture per protocol, across the dimensions the evidence supports.

Per-protocol posture

PROTOCOL	SESSIONS	BAND	SCORE	DIMENSIONS ASSESSED	NOT OBSERVABLE	ABSTENTIONS
smtp	1	CRITICAL	44	CERTIFICATE TRUST · CRYPTO CONFIGURATION · PROTOCOL VERSION	CERTIFICATE TRUST	2

6. Prioritised findings

Investigative priority, as ranked by the assessment. This ordering is not a severity ranking.

Severity states how serious a condition is. Priority states the order in which an analyst should work. The ML column shows any bounded adjustment the anomaly lane contributed to the ordering; it cannot move a finding across a severity tier and it never changes a severity.

Findings in investigative priority order

RANK	SEVERITY	ISSUE	STATUS	CERTAINTY	SESSIONS	PRIORITY	ML ADJ.
1	[!!] HIGH	Certificate key strength	OBSERVED ISSUE	CONFIRMED	1	84	0
2	[!!] HIGH	Certificate signature algorithm	OBSERVED ISSUE	CONFIRMED	1	84	0

7. Detailed findings

Certificate public key strength — A certificate uses an RSA key below the permitted length.

leaf uses a 1024-bit RSA key. NIST SP 800-57 Part 1 Rev.5 SS5.6.1: RSA moduli below 2048 bits provide less than 112 bits of security and are disallowed.

Certificate signature algorithm — A certificate in the chain is signed with a deprecated hash.

Signature algorithms observed: sha1WithRSAEncryption. SHA-1 -- RFC 9155 and NIST SP 800-131A Rev.2: SHA-1 must not be used for digital signature generation. A chain is only as sound as its weakest signature.

Recurrence counts the sessions in which a condition was observed. Findings are grouped by condition, so one misconfiguration seen many times is one issue with a recurrence count, not many issues.

Issue groups with evidence and standards basis

SEVERITY	ISSUE	FACT KIND	DIMENSION	CERTAINTY	OBSERVABILITY	RECURRENCE	STANDARD	FRAMES	PENALISING
[!!] HIGH	Certificate public key strength	Base security issue	Certificate trust	CONFIRMED	OBSERVABLE	1	UNMAPPED	6	Yes
[!!] HIGH	Certificate signature algorithm	Base security issue	Certificate trust	CONFIRMED	OBSERVABLE	1	UNMAPPED · UNMAPPED	6	Yes
[i] INFO	Certificate chain structure	Positive evidence	Certificate trust	CONFIRMED	—	1	UNMAPPED · UNMAPPED · UNMAPPED	—	No
[i] INFO	Certificate extraction	Positive evidence	Certificate trust	CONFIRMED	—	1	UNMAPPED · UNMAPPED	—	No
[i] INFO	Certificate validity period	Positive evidence	Certificate trust	CONFIRMED	—	1	UNMAPPED	—	No
[i] INFO	Negotiated TLS version posture	Positive evidence	Protocol version	CONFIRMED	—	1	NIST SP 800-52r2 · RFC 8996 (BCP 195)	—	No
[i] INFO	Forward secrecy	Positive evidence	Crypto configuration	CONFIRMED	—	1	NIST SP 800-52r2 · UNMAPPED	—	No
[i] INFO	Implicit TLS session	Positive evidence	Crypto configuration	UNCERTAIN	—	1	RFC 8314	—	No
[i] INFO	Insecure protocol configuration	Positive evidence	Crypto configuration	CONFIRMED	—	1	NIST SP 800-52r2 · RFC 8996 (BCP 195) · UNMAPPED · UNMAPPED	—	No
[i] INFO	Key exchange mechanism	Positive evidence	Crypto configuration	CONFIRMED	—	1	UNMAPPED	—	No

8. Standards basis

Standards cited by the rules that produced these findings.

Standards cited in this assessment

STANDARD	CITATIONS	SECTIONS
distinct_standards	3	—
note	every citation originates from a Phase-4/5 rule; the Phase-7 registry adds structure and introduces no new standard	—

A standard appearing here means a rule cited it as the basis for an observation. It is not a statement of compliance or certification against that standard.

9. AI / ML transparency

What the machine-learning lane did, and what it is not permitted to do.

ML lane	Disabled

ML lane disabled; posture computed from deterministic and cross-session evidence only

10. Abstentions and observability limits

Questions this assessment declined to answer, and what would settle them.

Abstentions

REASON	ISSUE CLASS	COULD NOT CONCLUDE	WHY	WOULD BE RESOLVED BY	PROTOCOL	FRAMES
Not observable	Certificate observability	Certificate trust and revocation boundary	Certificate trust and revocation were not evaluated.	not resolvable by passive capture; this fact is structurally hidden from an observer without key material	smtp	6
Insufficient capture	Tls handshake evidence	TLS handshake completion evidence	TLS negotiation was observed but completion is not established.	re-capture the full session including connection setup and teardown	smtp	4, 6, 8, 9, 10, 12

An abstention is neither a failure nor a pass. It records that the evidence did not support a conclusion. Treating any row above as compliant, or as a detected problem, misreads it.

11. Assessment limitations

Constraints that bound what this assessment can support.

passive analysis: conclusions describe what the capture shows, not the server's configuration

no attacker, intent or attribution is or can be established from a packet capture

certificate chain, expiry and key strength are not observable for TLS 1.3 or resumed sessions (RFC 8446 SS2, SS2.2)

absence of a STARTTLS advertisement is ambiguous: stripping and genuine non-support are byte-identical at the application layer

NyaySaakshya cannot confirm that any remediation was applied or effective

12. Evidence and provenance

What was analysed, by which engine, and on what evidence.

Assessment identifier	645090e0bc44ba9d
Capture identifier (SHA-256)	ce5377348e22ad92c33d705e31388944bad8224534d478c38b57a75d3ba4a501
Analysis run identifier	73ef94a35c4a442d8ed7d0f762175c8f
Analysis timestamp	2026-10-03T04:36:38Z
Posture schema version	1.0
Posture engine version	0.8.0
Report schema version	1.0
Renderer version	0.9.0

Every conclusion in this report traces to specific frames in the capture, to the rule that examined them, and to the standard that rule cited. The capture identifier above is the SHA-256 of the analysed bytes; a capture that hashes differently is a different capture and this assessment does not describe it.

Engine provenance

ITEM	VALUE
Contradictions recorded	0
Duplicate sources merged	0
Note	every severity, status and standard originates in a Phase-4 or Phase-5 rule; the posture engine aggregates and never re-judges
Rule ids	['SEC-CERT-001', 'SEC-CERT-002', 'SEC-CERT-003', 'SEC-CERT-004', 'SEC-CERT-005', 'SEC-CFG-001', 'SEC-FS-001', 'SEC-KEX-001', 'SEC-STLS-003', 'SEC-TLS-001', 'SEC-TLS-002', 'SEC-TLS-003']
Source counts	{'abstentions': 2, 'cross_session_findings': 0, 'deterministic_findings': 12, 'fused_findings': 10, 'ml_results': 0}
Versions	{'posture_engine': '0.8.0', 'posture_schema': '1.0', 'scoring_formula': 'F2-group-damped'}

13. Methodology

The stages that produced this assessment.

Input	Packet capture (PCAP)
Protocols assessed	smtp
Output	Canonical posture assessment

1. Capture validation and SHA-256 identity
2. Packet dissection
3. TCP stream and protocol session reconstruction
4. Deterministic security analysis against versioned, standards-bound rules
5. Cross-session reasoning
6. Evidence fusion, deduplication and contradiction handling
7. Risk classification, posture scoring, prioritisation and remediation

The machine-learning lane was not exercised for this assessment.

Report schema 1.0 · Renderer 0.9.0 · Posture schema 1.0 · Posture engine 0.8.0

The timestamp shown is the time of analysis, not of report generation. This report is a rendering of the canonical assessment and introduces no security conclusion of its own.